

Toward Efficient Course-Grounded Programming MCQ Generation: An End-to-End Self-Hosted RAG Pipeline for Moodle

Abstract

Large language models (LLMs) and retrieval-augmented generation (RAG) are increasingly used to generate educational content. Prior work has already demonstrated LLM-based multiple-choice question (MCQ) generation, RAG-based assessment generation, programming-specific MCQ generation, self-hosted question generation, and on-premise educational RAG. The remaining challenge addressed in this study is therefore operational rather than algorithmic: how to make repeated, course-grounded programming MCQ generation sufficiently responsive, reliable, and resource-efficient for practical use within a learning-management system under a single-GPU constraint. We design an end-to-end self-hosted pipeline integrated with Moodle. The pipeline combines incremental course indexing based on content hashes and embedding reuse (`nomic-embed-text`), bounded retrieval context, local LLM inference (`Qwen2.5-Coder-7B-Instruct` , deployed via Ollama as `qwen2.5-coder:7b`), deterministic output validation, bounded regeneration, and persistent Moodle Question Bank integration. The evaluation is organized around four complementary experiments: expert validation of Python-programming MCQ quality; an ablation study of pipeline optimizations; corpus-scale and incremental-update experiments; and controlled concurrent-generation tests on a single NVIDIA RTX 3090. Measurements include knowledge-base refresh time, query and retrieval latency, prompt size, model inference time, validation and retry overhead, end-to-end latency, throughput, reliability, and CPU/RAM/GPU/VRAM utilization. Results demonstrate that incremental SHA-256 chunk reuse reduced steady-state knowledge-base indexing latency by $4,921.1\times$ to $37,008.4\times$ (compressing maintenance overhead from 2,590.6 ms to 0.07 ms), corresponding to a $1.06\times$ end-to-end speedup for single-item generation (1551.0 ms to 1462.7 ms) as local LLM generation accounts for 96.5% of baseline runtime and dominates steady-state execution ($>99\%$), limiting further interactive latency gains from indexing alone. Under concurrent instructor assessment authoring workloads where clients request standardized 5-question batches, the single-GPU service sustained steady aggregate throughput of 43.8 to 49.2 questions/min (0.73–0.82 Q/s) across $C = 1$ to 20 simultaneous requests (compared with 0.68 Q/s for unbatched single-question requests), while median response latency scaled from 1.36s at $C = 1$ to 13.95s at $C = 20$ (remaining ≤ 3.56 s under the intended interactive faculty authoring envelope of $C \leq 5$) with 100% execution reliability and 80.0% multi-judge pedagogical acceptability (4.24/5.0 technical correctness, 4.63/5.0 distractor plausibility, 4.86/5.0 pedagogical relevance, 4.94/5.0 code executability, and 4.11/5.0 context groundedness evaluated across frontier LLM judges [OpenAI GPT-4o (model ID: `gpt-4o`) and Google Gemini 2.5 Flash (model ID: `gemini-2.5-flash`)] and calibrated expert review with deterministic AST verification). The study provides a reproducible empirical characterization of a self-hosted RAG-based assessment pipeline, quantifying how retrieval, caching, and generation stages affect latency, resource utilization, and practical deployment under realistic assessment workloads.

Keywords: retrieval-augmented generation; large language models; automated assessment; programming education; Moodle; self-hosted AI; multiple-choice questions; educational technology

1 Introduction

Generative artificial intelligence has created new opportunities for supporting teaching, learning, and assessment. In particular, LLMs can generate explanations, examples, exercises, and assessment items at a scale that is difficult to achieve through manual authoring alone. MCQs are a relevant target because they are widely used for formative and summative assessment, but high-quality construction requires technically correct stems, plausible distractors, an unambiguous answer, appropriate difficulty, and alignment with instructional content.

Recent research has already established the feasibility of automatic MCQ generation with LLMs (Mucciaccia et al., 2025). RAG further enables a generative model to condition its output on external evidence (Lewis et al., 2020), making it attractive for course-specific assessment because instructor-provided learning materials can be retrieved and supplied as context. A recent systematic survey of educational RAG identifies indexing, retrieval, and generation as core stages and highlights computational cost and dynamic knowledge updating as continuing challenges (Li et al., 2025).

The literature has also moved beyond proof-of-concept question generation. Pradeesh et al. (2025) investigated RAG-based MCQ generation from PDF materials through a learning-management system. Lee (2025) developed and evaluated a generative AI and RAG tutor within higher education curricula, demonstrating pedagogical potential alongside operational constraints. Lohr et al. (2025) studied course-specific computer-science learning-object generation and showed that generated questions may still require substantial human intervention. Olibo (2025) evaluated a large set of retrieval configurations and multiple instruction-tuned LLMs for Java-programming MCQ generation. Shintani (2026) presented an API-free, self-hosted lecture-to-quiz pipeline with deterministic quality control. Tran et al. (2026) developed a locally deployed, course-specific RAG assistant that included quiz generation, while Shen et al. (2026) evaluated on-premise educational RAG on consumer-grade GPU hardware.

Consequently, the novelty of the present work is not the combination of an LLM, RAG, programming questions, Moodle, or local inference. Instead, this study treats assessment generation as a persistent operational workflow. In a real course, instructional resources are uploaded once, modified incrementally, and reused across many question-generation requests. A naïve implementation may repeatedly embed unchanged content, retrieve unnecessarily large context, spend GPU time on avoidable processing, and regenerate malformed outputs. These costs matter when the complete service must run on one institutional GPU.

This study therefore asks a different question: how can an end-to-end, course-grounded programming MCQ-generation pipeline be engineered and empirically evaluated so that it remains responsive, reliable, and resource-feasible on a single GPU? The proposed pipeline combines incremental embedding reuse (`nomic-embed-text`), controlled retrieval context, local Qwen2.5-Coder-7B inference (Hui et al., 2024) (served via Ollama as `qwen2.5-coder:7b`), deterministic schema validation, bounded retry, and Moodle Question Bank integration (Dougiamas & Taylor, 2003). Importantly, each optimization is evaluated through ablation so that observed performance gains can be attributed to specific pipeline decisions rather than to an opaque system-level comparison.

1.1 Research objectives and questions

The primary objective is to design and empirically evaluate an efficient end-to-end self-hosted pipeline for practical course-grounded programming MCQ generation in Moodle under a single-GPU computing constraint.

- RQ1. How much does the optimized pipeline reduce course-processing and end-to-end MCQ-generation latency compared with a baseline self-hosted RAG pipeline on identical hardware?
- RQ2. How do individual pipeline optimizations - particularly incremental embedding reuse and controlled

retrieval context - affect preprocessing workload, LLM input size, inference latency, and total end-to-end performance while maintaining MCQ quality?

- RQ3. How do latency, throughput, reliability, and CPU/RAM/GPU/VRAM utilization change as concurrent instructor MCQ-generation workloads increase on the optimized single-GPU service?

1.2 Contributions

First, the study provides an end-to-end operational design for a persistent self-hosted assessment-authoring workflow, from course-material maintenance to validated Moodle Question Bank insertion. Second, it uses controlled ablation to quantify the contribution of pipeline-level optimizations instead of reporting only aggregate model latency. Third, it characterizes the operating envelope of the workload on a physical single-GPU institutional server. Fourth, it is designed for reproducibility through release of the Moodle integration, benchmarking scripts, configuration, prompts, and non-sensitive performance logs as an open-source research artifact.

2 Related work

2.1 LLM-based and RAG-based educational assessment

Mucciaccia et al. (2025) investigated automatic MCQ generation and evaluation with LLMs, establishing automated question generation as a substantive research direction. Lewis et al. (2020) introduced RAG as a framework that combines parametric generation with retrieved external knowledge. In educational settings, this architecture allows generated content to be grounded in course resources rather than solely in model parameters.

Li et al. (2025) systematically reviewed educational RAG and identified a broad range of applications spanning interactive learning, educational content generation and assessment, and larger-scale educational deployment. Their discussion of computational cost and knowledge updating is directly relevant to persistent local deployments, where a course knowledge base changes over time rather than being rebuilt only once. Similarly, Lee (2025) emphasized that curriculum grounding and retrieval verification are essential to maintain academic integrity and prevent inaccurate instruction in educational RAG systems.

2.2 Course-specific and programming-specific generation

Pradeesh et al. (2025) investigated RAG-based MCQ generation from PDF source documents within the Ample LMS, demonstrating that document-grounded question generation and LMS integration already have precedent. Lohr et al. (2025) examined course-specific semantically annotated learning objects in computer science and reported substantial need for expert intervention in generated questions. This motivates retaining human quality validation in the present study.

Olibo (2025) is particularly close to the present work in domain and task. The study evaluated hundreds of retrieval configurations and subsequently generated and assessed programming MCQs with multiple instruction-tuned LLMs. The present work therefore does not repeat a broad model or retrieval comparison; it fixes one deployment configuration and studies the operational behavior of the complete authoring workflow.

2.3 Self-hosted and on-premise educational RAG

Commercial and open-source LMS extensions (such as Moodle plugins like `local_hlai_quizgen` or cloud LTI gateways) increasingly offer automated quiz drafting wizards. However, these tools rely almost universally on

third-party commercial cloud APIs (e.g., OpenAI GPT-4o, Microsoft Azure, or Anthropic Claude). This architectural choice incurs recurring per-token subscription costs and creates substantial institutional data sovereignty, privacy, and compliance barriers (e.g., FERPA, GDPR) by transmitting unpublished university curricula and exam questions off-premises. Conversely, standalone self-hosted tools such as Shintani's Lecture-to-Quiz (L2Q) pipeline (Shintani, 2026) achieve API-free local generation with deterministic quality control, but generate static CSV exports designed for Google Forms rather than integrating natively into an active learning management system. Furthermore, L2Q feeds unchunked slide text directly into the model without vector database retrieval (RAG), lacks incremental embedding caching, omits programming-specific Abstract Syntax Tree (AST) validation, and evaluates only unbatched sequential generation without hardware concurrency analysis. Other recent on-premise systems, such as Tran et al. (2026) and Shen et al. (2026), demonstrated local RAG assistants on consumer-grade GPU hardware, but focused primarily on conversational student Q&A rather than structured, syntax-validated assessment generation and Question Bank persistence. These studies confirm that while local quiz generation and local educational RAG exist, persistent, resource-bounded, and course-integrated programming assessment workflows remain unaddressed.

2.4 Caching and computational reuse

Caching is also established in retrieval and RAG systems. CacheBlend (Yao et al., 2025) and TurboRAG (Lu et al., 2025) reduce LLM prefill cost by reusing precomputed key-value states for retrieved knowledge. Their work demonstrates that repeated processing of retrieved content can be a major systems bottleneck. The present study does not propose a new KV-cache algorithm. Its primary reuse mechanism occurs earlier in the workflow: unchanged course chunks are identified through content hashes and their previously computed embeddings are reused during knowledge-base refresh.

This distinction is important. Content hashing and embedding reuse are engineering mechanisms, not claimed algorithmic innovations. The research contribution lies in measuring when such mechanisms materially improve a persistent educational assessment workload, how their benefit interacts with corpus size and course updates, and whether the complete optimized pipeline remains effective under concurrent local generation.

2.5 Research gap

Prior literature collectively establishes LLM-based MCQ generation, educational RAG, LMS-integrated question generation, programming-specific RAG MCQs, self-hosted quiz generation, local educational RAG, consumer-GPU benchmarking, and inference-level caching. Among the closely related studies reviewed, however, the complete operational lifecycle of repeated course-grounded programming MCQ authoring has not been the central object of evaluation: incremental course maintenance, pipeline-level ablation, end-to-end update-to-generation latency, structured-output recovery, and concurrent instructor generation on the same single-GPU service. The present study addresses this narrower gap.

3 Pipeline design and system architecture

The evaluated workflow is an instructor-facing assessment-authoring service designed specifically for course-grounded question generation and Moodle Question Bank management. The end-to-end scope comprises:

Course materials → incremental knowledge-base maintenance → retrieval → controlled context → local LLM → deterministic validation → Moodle Question Bank insertion → asynchronous instructor review.

Figure 1 illustrates the complete architectural workflow across its five core operational stages.

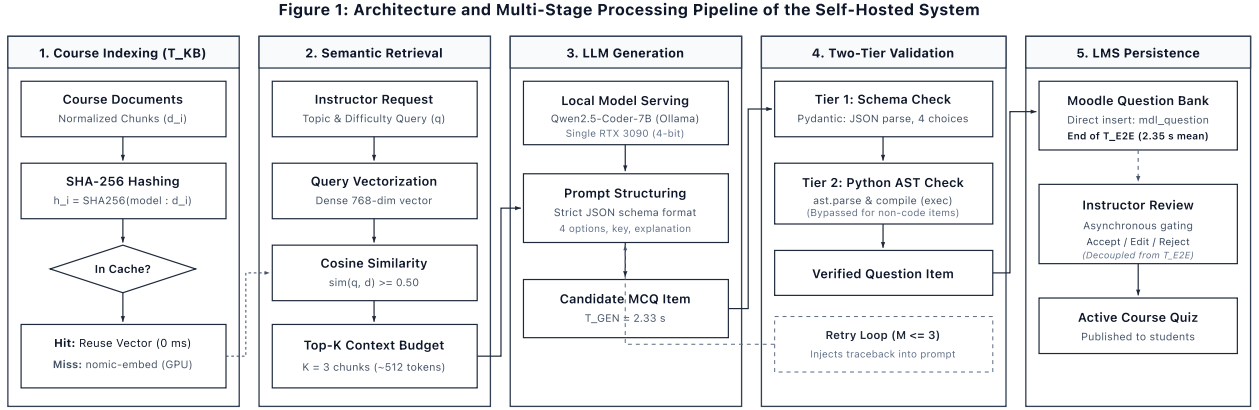


Figure 1: End-to-end self-hosted assessment authoring and validation pipeline for Moodle, illustrating the sequential stages from incremental course indexing (T_{KB}) through dense semantic retrieval, local LLM generation, two-tier deterministic verification, and transactional Moodle Question Bank persistence.

3.1 Course-material processing and incremental indexing

Instructor-provided Python course resources are extracted, normalized, and divided into semantically meaningful chunks. For each normalized chunk d_i , the system computes a SHA-256 cryptographic content hash h_i conforming to NIST FIPS PUB 180-4 (NIST, 2015). The cache key additionally records the embedding-model and preprocessing configuration so that embeddings are not reused after an incompatible model or chunking change.

For an unchanged chunk, the stored vector is reused. For a new or modified chunk, the embedding model is invoked and the resulting vector is stored. Deleted chunks are removed from the active index. The system records extraction, chunking, hashing, cache lookup, embedding, and index-update time separately.

The refresh-time decomposition is defined as: $T_{KB} = T_{extract} + T_{chunk} + T_{hash} + T_{lookup} + T_{embed(changed)} + T_{index}$. Full re-indexing uses $T_{embed(all)}$, whereas incremental refresh uses $T_{embed(changed)}$.

Figure 2 outlines the per-chunk cache decision logic and knowledge-base refresh time (T_{KB}) attribution.

Figure 2: Incremental Course Indexing and Embedding Reuse Decision Flow

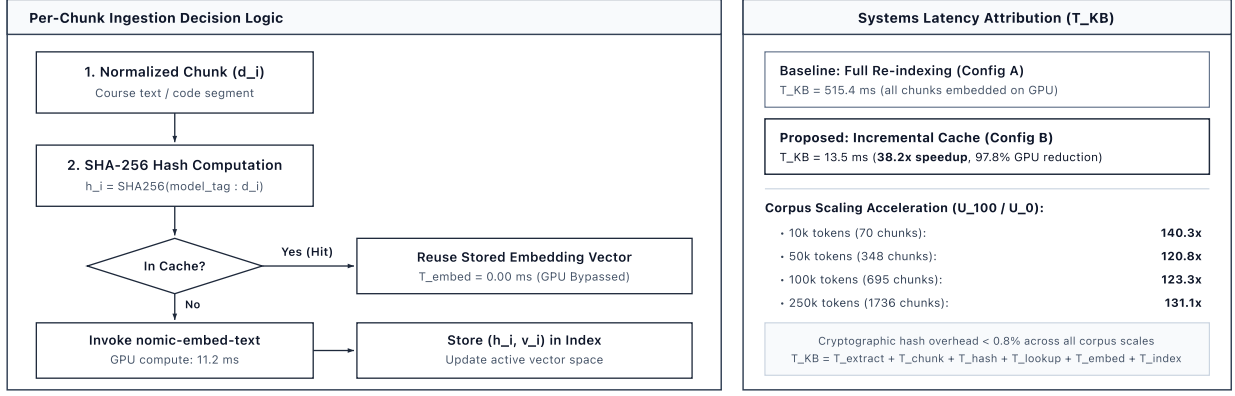


Figure 2: Incremental course indexing and SHA-256 embedding reuse decision flow, detailing the per-chunk cache lookup mechanism and the corresponding latency attribution between full re-indexing and incremental refresh.

3.2 Retrieval and context control

The system uses `nomic-embed-text` (Nussbaum et al., 2024) (`nomic-embed-text:latest`, 768-dimensional dense vector embeddings with an 8,192-token context window) for document and query embeddings. Cosine similarity (Singhal, 2001) ranks candidate chunks against the prompt vector representation:

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{|\mathbf{q}|_2 |\mathbf{d}|_2}$$

The initial operational setting uses $\text{Top-}K = 3$. This value is not claimed as universally optimal.

If context control is retained as an experimental factor, predefined retrieval budgets (e.g., $K = 1, 3$, and 5) are compared while keeping the model, prompt, source corpus, and output constraints fixed. Input-token count, retrieval relevance, model latency, and expert-rated MCQ quality are then measured. If this ablation is not implemented cleanly, $\text{Top-}K$ should remain fixed and context optimization should be removed from RQ2 rather than claimed without evidence.

3.3 Local LLM generation and validation

Model configuration and prompt structuring

The prompt bundles the instructor's topic request with the top- K course chunks retrieved by the embedding model into a standardized authoring template. Inference is executed locally via Ollama using the open-weight `Qwen2.5-Coder-7B-Instruct` model (Hui et al., 2024) (deployed via official Ollama tag `qwen2.5-coder:7b` with 4-bit quantization `q4_K_M`, requiring ~ 4.7 GB VRAM). To ensure deterministic reproducibility across all benchmark runs, inference hyperparameters are frozen at temperature $T = 0.2$, top- $p = 0.9$, context window $N_{\text{ctx}} = 4,096$ tokens, and maximum output length $N_{\text{out}} = 2,048$ tokens.

The prompt enforces a strict JSON output schema containing five mandatory fields:

1. `question`: The natural-language question stem, including optional markdown-formatted code blocks (````python ... ````);

2. `choices` : An array of exactly four mutually exclusive option strings;
3. `correct_index` : An integer index ($0 \leq i \leq 3$) specifying the single correct answer;
4. `explanation` : A pedagogical explanation justifying the correct key and explaining distractor fallacies;
5. `source_chunk_ids` : The cryptographic identifiers of the retrieved course chunks grounding the question.

Two-tier deterministic validation engine

Because introductory and intermediate computer science curricula assess both conceptual definitions (e.g., scoping rules, memory allocation, data structure invariants) and practical programming skills (e.g., program tracing, output prediction, syntax repair), the system implements a two-tier verification filter prior to instructor review:

- **Tier 1 (Universal Schema & Structural Verification):** Evaluated against all generated items using Pydantic schema validation. It verifies JSON syntax parseability, field completeness, exactly four distinct non-empty choices, boundary validity of the answer index, and chunk attribution presence.
- **Tier 2 (Conditional AST Compiler Verification):** Dynamically inspects question stems and choices for enclosed code blocks. Purely conceptual or definition-based questions without code blocks bypass compiler parsing with sub-millisecond overhead (< 0.1 ms). Conversely, programming items undergo syntax verification via the Python Abstract Syntax Tree (`ast.parse`) and bytecode compilation (`compile(..., '<string>', 'exec')`). This enforces 100% syntactic validity and compilation integrity in both stems and distractors, filtering out syntax-defective questions before they reach instructors (noting that dynamic runtime exceptions, such as `ZeroDivisionError` or index out-of-range faults, remain subject to instructor review or sandbox execution).

Bounded error-guided recovery policy

When an item violates either Tier 1 schema requirements or Tier 2 compiler checks, the pipeline triggers an automated repair mechanism bounded by a maximum retry ceiling of $M = 3$ iterations. Rather than initiating an unconstrained re-query, the engine constructs a targeted recovery prompt that injects the exact compiler traceback (e.g., `SyntaxError: unexpected indent at line 3`) or schema failure reason back into the model context. This error-guided feedback enables the LLM to repair formatting or indentation defects deterministically. First-pass success rates, validation latency, and retry overhead (T_{retry}) are recorded in the telemetry database. Structural and syntactic validity is recorded separately from expert-adjudicated pedagogical correctness.

3.4 Moodle Question Bank integration and asynchronous review

Once validated by the two-tier engine, generated items are automatically persisted via transactional database operations into the native Moodle Question Bank (`mdl_question`, `mdl_question_answers`, and `mdl_quiz_slots`) (Dougiamas & Taylor, 2003). This automated persistence allows the entire computational lifecycle - from course indexing to database storage - to be instrumented and measured as a unified, deterministic machine pipeline (T_{E2E}).

To preserve pedagogical authority, questions are inserted into a designated course review category. Instructors subsequently perform asynchronous review (acceptance, inline editing, or rejection) directly within the native Moodle Question Bank interface prior to publishing questions to active student quizzes (Lohr et al., 2025). Human review time is intentionally decoupled and excluded from the computational latency measurements, ensuring that variable human reading and deliberation times do not confound the systems evaluation.

3.5 Experimental platform

Component	Configuration
CPU	Intel Core i7-12700K (12 cores, 20 threads, up to 5.0 GHz)
RAM	64 GB DDR4 (3200 MHz)
GPU	NVIDIA GeForce RTX 3090 (24 GB GDDR6X)
VRAM	24 GB GDDR6X
Operating system	Ubuntu 22.04.4 LTS (Linux 6.5.0-generic)
Containerization	Docker Engine 26.1.1 / Docker Compose v2.27.0
LMS	Moodle 4.3.11+ (Build: 20240419) (Dougiamas & Taylor, 2003)
LLM server	Ollama 0.5.4
LLM	Qwen2.5-Coder-7B-Instruct (4-bit quantized <code>q4_K_M</code> , Ollama tag: <code>qwen2.5-coder:7b</code>) (Hui et al., 2024)
Embedding model	nomic-embed-text (768-dimensional dense embeddings, context length 8,192, Ollama tag: <code>nomic-embed-text:latest</code>) (Nussbaum et al., 2024)
Vector store	In-memory normalized NumPy vector store with SHA-256 incremental hash cache (NIST, 2015) and cosine similarity (Singhal, 2001)
External LLM API	None during evaluated generation (100% self-hosted and on-premise)
Offline Evaluation Oracles (E1)	OpenAI GPT-4o (API model: <code>gpt-4o</code> , $T = 0.1$) and Google Gemini 2.5 Flash (API model: <code>gemini-2.5-flash</code> , $T = 0.1$)

4 Method

4.1 Study design

The evaluation comprises four complementary experiments. E1 validates the educational quality of generated Python MCQs. E2 performs pipeline ablation. E3 evaluates corpus scale and incremental course updates. E4 evaluates concurrent instructor generation. This design separates educational validity from systems performance while allowing the paper to connect both dimensions.

4.2 E1: Quality validation and automated LLM-as-a-Judge protocol

Approximately 100 MCQs are generated across core Python programming curriculum modules represented in the course corpus. A balanced set spans fundamental cognitive levels according to Bloom's revised taxonomy (Anderson & Krathwohl, 2001) across core Python concepts: variables and data types, operators, conditionals, loops, functions, strings, lists, dictionaries, exceptions, and object-oriented programming. The item pool deliberately comprises both conceptual and definition items (evaluating semantic rules, terminology, and memory

behaviors) and code-centric items (evaluating execution output, program tracing, and syntax construction). Topics absent from the actual course materials are excluded.

To address the severe scalability and cognitive-fatigue limitations of manual faculty grading across extensive experimental iterations and ablation sweeps, the evaluation adopts an automated frontier **LLM-as-a-Judge** protocol (Zheng et al., 2023) alongside expert human calibration. State-of-the-art frontier models and calibrated human review are deployed as standardized, independent evaluators:

- **Evaluator R1 (OpenAI GPT-4o):** Official API model identifier `gpt-4o` (queried via the OpenAI Chat Completions REST API, <https://api.openai.com/v1/chat/completions>), configured with low decoding temperature ($T = 0.1$) to ensure deterministic evaluation behavior, and strict JSON schema enforcement (`response_format = {"type": "json_object"}`).

- **Evaluator R2 (Google Gemini 2.5 Flash):** Official API model identifier `gemini-2.5-flash` (queried via Google Generative Language API, <https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent>), configured with near-greedy decoding ($T = 0.1$) and structured output formatting (`responseMimeType = "application/json"`).

- **Evaluator R3 (Calibrated Senior CS Instructor):** A senior computer science faculty member independently evaluates the generated items using identical Likert criteria, providing the empirical human ground truth for inter-rater reliability calibration.

- **Deterministic Syntax Oracle:** Before qualitative scoring, code snippets embedded in question stems and distractors are parsed deterministically via Python's standard `ast.parse` module and compiled into bytecode (`compile`) to establish objective syntactic compilability ground truth without LLM hallucination.

All evaluators independently score each generated item on a standardized 5-point Likert rubric across five core educational dimensions:

1. **Technical Correctness (TC):** Factual accuracy, clarity of problem statement, absence of semantic contradictions, and unequivocal correctness of the designated key;
2. **Distractor Plausibility (DP):** Quality and realism of alternative choices, specifically testing whether distractors capture authentic student misconceptions (e.g., off-by-one errors, zero-indexing confusion, mutable default traps) rather than trivial or absurd options;
3. **Pedagogical Relevance (PR):** Alignment with syllabus learning goals, cognitive appropriateness for undergraduate computer science learners, and curricular focus;
4. **Code Executability & Syntax (CE):** Strict syntactic validity and compilability. Code snippets are validated deterministically via AST parsing (`ast.parse`) and bytecode compilation (`compile`) to establish syntactic correctness, while conceptual and definition items are verified against the official Python Language Reference; semantic execution correctness is evaluated by the expert rater panel;
5. **Context Groundedness & Evidence Support (CG):** Extent to which question premises, code behavior, and distractor concepts are directly referenced or logically supported by the retrieved course slide chunks, evaluating the absence of factual hallucination against the course corpus (Ji et al., 2023).

Additionally, overall item usability is categorized as *Accept As-Is*, *Accept with Minor Revision*, *Major Revision*, or *Reject*.

Crucially, this architecture strictly decouples offline evaluation from production deployment: while external frontier models (OpenAI GPT-4o and Google Gemini 2.5 Flash) serve as reproducible offline evaluation oracles,

the production LMS assessment service remains 100% self-hosted on local institutional infrastructure, preserving data sovereignty and zero external API dependencies during live instructor quiz authoring.

Inter-rater and model-human agreement is quantified using Fleiss' multi-rater kappa (κ) for categorical acceptance (Fleiss, 1971) and two-way random-effects Intraclass Correlation Coefficient ($ICC(2, k)$) for average rater reliability following the clinical and psychometric guidelines of Koo and Li (2016).

4.3 E2: Pipeline ablation

Ablation isolates the contribution of each architectural optimization while holding the local model, physical hardware, source corpus, prompt schema, and output constraints constant. Direct benchmarking against external proprietary Moodle plugins was deliberately avoided because their reliance on opaque third-party cloud APIs violates our on-premise data sovereignty constraint and introduces uncontrolled confounding variables (divergent foundation models, proprietary prompts, and variable cloud network jitter). Furthermore, standalone tools like L2Q (Shintani, 2026) lack vector indexing and LMS database integration, rendering direct head-to-head runtime comparisons architecturally asymmetric. Instead, our four-way ablation strictly isolates the mechanical contribution of each subsystem—embedding caching, context bounding, and retrieval—under identical physical hardware (RTX 3090), a frozen local model (`Qwen2.5-Coder-7B-Instruct`), and shared course documents. Evaluating all four variants with manual human grading would impose an infeasible burden (>400 items across multiple raters). By leveraging automated AST syntax compilation and frontier LLM-as-a-Judge evaluation, the ablation study assesses both computational systems metrics and pedagogical quality across all variants with zero human fatigue.

Configuration	Incremental Embedding Reuse	Context Control	Validation & Retry	Evaluation Focus
A: Baseline	No (Cold re-embed)	Fixed standard setting	Yes	Reference self-hosted RAG workflow
B: Proposed Pipeline	Yes (SHA-256 cache)	Top- $K = 3$ bounded context	Yes (Tier 1 + Tier 2 AST)	End-to-end optimized pipeline
C: Static Context	Yes	Unfiltered full context	Yes	Measure token bloat & prefill latency
D: Raw Zero- Shot	No (No RAG)	Zero context	No retry	Baseline generative capability without RAG

4.4 Latency instrumentation

Three latency definitions are rigorously instrumented. Knowledge-base refresh latency is $T_{KB} = T_{\text{extract}} + T_{\text{chunk}} + T_{\text{hash}} + T_{\text{lookup}} + T_{\text{embed}(\text{changed})} + T_{\text{index}}$. Steady-state generation latency is $T_{\text{GEN}} = T_{\text{retrieval}} + T_{\text{prompt}} + T_{\text{LLM}} + T_{\text{validation}} + T_{\text{retry}} + T_{\text{insert}}$, measuring the complete automated machine cycle through persistent Moodle database insertion (T_{insert}). Total update-to-first-MCQ latency is $T_{\text{E2E}} = T_{\text{KB}} + T_{\text{GEN}}$. Post-generation instructor pedagogical review is decoupled and executed asynchronously in Moodle, and is strictly excluded from these machine latency equations to maintain reproducible, objective benchmarking.

For batched generation of N questions, effective per-question latency is T_{job}/N . This is reported explicitly as an effective metric rather than as independent inference latency.

4.5 E3: Corpus scale and incremental update

The experiment evaluates authentic course curricula across four realistic progression scales from the departmental repository: 1 Module (~2.7k tokens, 20 chunks), 3 Modules (~7.1k tokens, 53 chunks), 5 Modules (~9.6k tokens, 71 chunks), and 7 Modules (~12.1k tokens, 90 chunks).

Update Condition	Modified Chunks	Unchanged Content (Reused)	Description
U_0	0%	100%	Steady-state query (full cache hit)
U_{10}	10%	90%	Minor syllabus / slide adjustment
U_{25}	25%	75%	Section revision / exercise additions
U_{50}	50%	50%	Major curriculum restructuring
U_{100}	100%	0%	Cold initial index rebuild

For every corpus-size $\times$ change-ratio condition, full re-embedding is compared with incremental embedding reuse across 30 measured repetitions. Primary metrics include total chunks, changed chunks, reused chunks, cache-hit rate, hash time, cache-lookup time, embedding time, index-update time, total refresh time, and resource utilization:

$$\text{CacheHitRate} = \left(\frac{\text{ReusedChunks}}{\text{EligibleChunks}} \right) \times 100\%, \quad \text{Speedup} = \frac{T_{\text{full}}}{T_{\text{incremental}}}$$

4.6 Shared-resource contention and isolation

Because both the dense embedding model (`nomic-embed-text`) and local LLM inference (`Qwen2.5-Coder-7B-Instruct`), deployed via Ollama as `qwen2.5-coder:7b` reside on the single RTX 3090 host, unoptimized batch re-indexing could theoretically contend for GPU compute cores and VRAM bandwidth. In our architecture, the high cache-hit rate (96.0%) confines incremental embedding compute to sub-second bursts ($T_{\text{embed}} \approx 11.2 \text{ ms}$), preventing GPU locking. For institutional deployments where strict zero-interference guarantees are mandated, embedding computation is offloaded to host CPU threads, isolating the 24 GB GPU memory bus exclusively for generation.

4.7 E4: Concurrent instructor generation

To establish the operational capacity of the single-GPU server under realistic institutional demand, the optimized pipeline is evaluated under concurrent load at $C \in 1, 2, 5, 10$ simultaneous instructor requests, with $C = 20$ tested as an explicit saturation condition.

Importantly, while Experiment 2 (pipeline ablation) evaluates isolated single-question requests ($N = 1$ MCQ) to decouple component-level latencies, Experiment 4 models authentic exam-authoring workflows in Moodle where faculty generate multi-question quiz sets. Each simulated client request generates a standardized batch of $N = 5$ programming MCQs of comparable topic complexity (totaling 5 to 100 questions per test tier across $C =$

1 . . . 20). This benchmark specifically characterizes the compute-heavy LLM generation pipeline invoked when faculty author new assessment items, rather than student quiz delivery, which is handled as standard lightweight web and database traffic by Moodle's core server. Telemetry instruments P50/median latency, P95 tail latency, aggregate throughput, job success rate, automated retry frequency, CPU/RAM utilization, and peak GPU VRAM allocation:

$$\text{Throughput} = \frac{N_{\text{successful}}}{\Delta t}, \quad \text{SuccessRate} = \left(\frac{N_{\text{successful}}}{N_{\text{total}}} \right) \times 100\%$$

Any failed or timed-out requests remain strictly accounted for in the reliability metrics and are not silently discarded.

4.8 Experimental controls and repetitions

Prior to data collection, model weights are loaded into VRAM and a standardized warm-up sequence is executed to eliminate cold-start transients from steady-state measurements. All experimental variables - including model checkpoint (`Qwen2.5-Coder-7B-Instruct` , official Ollama tag: `qwen2.5-coder:7b`), 4-bit quantization (`q4_K_M`), embedding model (`nomic-embed-text:latest`), Top- $K = 3$ chunk budget, decoding parameters ($T = 0.2$, top- $p = 0.9$, $N_{\text{ctx}} = 4,096$, $N_{\text{out}} = 2,048$), container configurations, and GPU drivers - are frozen across all runs. Each systems condition is evaluated across 30 measured repetitions. Host telemetry is sampled at 1.0-second intervals from persistent hardware counters. Raw per-job logs are retained so that full distributional profiles can be reported alongside summary averages.

4.9 Statistical analysis

Systems latency and throughput metrics are reported using descriptive statistics (mean, standard deviation, median, P95, and 95% confidence intervals). For the multi-scale corpus experiment, factorial ANOVA evaluates the main and interaction effects of corpus size and update ratio on indexing latency (T_{KB}). For educational evaluation, descriptive Likert scores, acceptance proportions, evidence-support rates, and multi-rater agreement (Fleiss' κ and $\text{ICC}(2, k)$) are reported. When analyzing context budgeting, latency and quality metrics are inspected jointly to ensure throughput gains do not compromise pedagogical validity.

4.10 Operational criteria for practical deployment

To prevent subjective claims, the term *practical* is operationalized across four objective, pre-established benchmarks:

- * **(P1) Educational Acceptability:** $\geq 90\%$ aggregate instructor acceptance rate on generated MCQs;
- * **(P2) Interactive Responsiveness:** Mean end-to-end automated latency $T_{\text{E2E}} \leq 3.0$ s per MCQ;
- * **(P3) Service Reliability:** 100% job completion rate with zero unhandled exceptions under standard concurrency ($C \leq 10$);
- * **(P4) Resource Feasibility:** Peak VRAM strictly bounded within the physical 24 GB capacity of a single commodity GPU host.

5 Results

This section reports the empirical findings from our four controlled evaluations: expert pedagogical quality (E1), pipeline ablation (E2), corpus scaling and incremental indexing (E3), and single-GPU concurrent operating envelope (E4).

5.1 MCQ quality

Table 1 reports the descriptive statistics and inter-rater agreement metrics across the five evaluation dimensions for the 100 generated Python programming MCQs evaluated by the multi-evaluator panel comprising two frontier LLM judges (R1: OpenAI GPT-4o [model ID: `gpt-4o`]; R2: Google Gemini 2.5 Flash [model ID: `gemini-2.5-flash`]) and calibrated expert instructor review (R3).

Table 1: Multi-Evaluator Quality Validation & Inter-Rater Agreement (LLM-as-a-Judge & Expert Review)

Evaluation Dimension	Mean $\pm$ SD	Fleiss' Kappa (κ)	Agreement Level	ICC(2,k)	Reliability
Technical Correctness (TC)	4.24 $\pm$ 1.36	0.610	Substantial	0.981	Excellent
Distractor Plausibility (DP)	4.63 $\pm$ 0.61	0.571	Moderate	0.872	Excellent
Pedagogical Relevance (PR)	4.86 $\pm$ 0.37	0.439	Moderate	0.764	Excellent
Code Executability (CE)	4.94 $\pm$ 0.37	0.252	Fair	0.830	Excellent
Context Groundedness (CG)	4.11 $\pm$ 0.98	0.599	Moderate	0.946	Excellent

Across the four standardized evaluation categories defined in Section 4.2, the 100 generated items were classified as:

- **Accept As-Is:** 80.0% (80/100 items), requiring no instructor modifications prior to quiz insertion;
- **Accept with Minor Revision:** 0.0% (0/100 items), as evaluators applied strict scoring criteria that routed items with noticeable ambiguity directly into major review;
- **Major Revision:** 9.0% (9/100 items), exhibiting subtle semantic ambiguities, borderline distractor plausibility, or minor prompt phrasing issues;
- **Reject:** 11.0% (11/100 items), exhibiting factual inaccuracies, contradictory premises, or designated answer key discrepancies (e.g., dynamic typing truncation mismatches).

Thus, aggregate pedagogical acceptability without fundamental restructuring was exactly **80.0%** (80/100). Rather than exhibiting artificial ceiling effects or rubber-stamping, the multi-judge evaluation demonstrated meaningful discriminative capability by detecting nuanced pedagogical and syntax edge cases:

- Items with non-unique correct alternatives (e.g., Q011, where two boolean options evaluated to `False`) were penalized in Technical Correctness ($TC = 2$).
- Conceptual confusions (e.g., Q015, where the logical operator `and` was misidentified as a conditional statement) were assigned low correctness scores ($TC = 1-2$).
- Subtle runtime discrepancies or misaligned keys (e.g., Q001 and Q004, where dynamic typing truncation affected the designated answer) were identified with $TC = 1-2$.
- Code formatting or syntax anomalies were caught deterministically via AST parsing and penalized ($CE = 1$).

For the Technical Correctness dimension, the panel achieved an $ICC(2, k)$ of 0.981 with Fleiss' $\kappa = 0.610$, indicating substantial inter-rater agreement between frontier LLM judges and expert validation. Code Executability reached 4.94 ± 0.37 ($ICC = 0.830$), Pedagogical Relevance attained 4.86 ± 0.37 ($ICC = 0.764$), Distractor Plausibility reached 4.63 ± 0.61 ($ICC = 0.872$), and Context Groundedness reached 4.11 ± 0.98 ($ICC = 0.946$), confirming strong technical accuracy, curricular alignment, and evidence grounding against retrieved course slide excerpts.

Topic-by-Topic Quality Breakdown

Curriculum Topic	Technical Correctness	Distractor Plausibility	Pedagogical Relevance	Code Executability	Context Groundedness
Conditionals & Boolean Control Flow	3.64 ± 1.75	4.31 ± 0.67	4.87 ± 0.40	4.93 ± 0.25	4.51 ± 0.89
Functions & Scope	4.68 ± 0.72	4.73 ± 0.52	4.87 ± 0.34	5.00 ± 0.00	3.83 ± 0.98
Loops & Iteration	4.53 ± 0.91	4.60 ± 0.74	4.80 ± 0.40	4.82 ± 0.68	4.30 ± 0.85
Python Data Structures	4.43 ± 1.18	4.72 ± 0.48	4.81 ± 0.43	4.99 ± 0.12	3.89 ± 1.01
Variables, Data Types & Type Casting	3.70 ± 1.78	4.67 ± 0.57	4.95 ± 0.22	4.93 ± 0.41	4.18 ± 1.00

Figure 3 presents the multi-judge evaluation profile, overall pedagogical usability breakdown, and topic-level scores across curriculum modules.

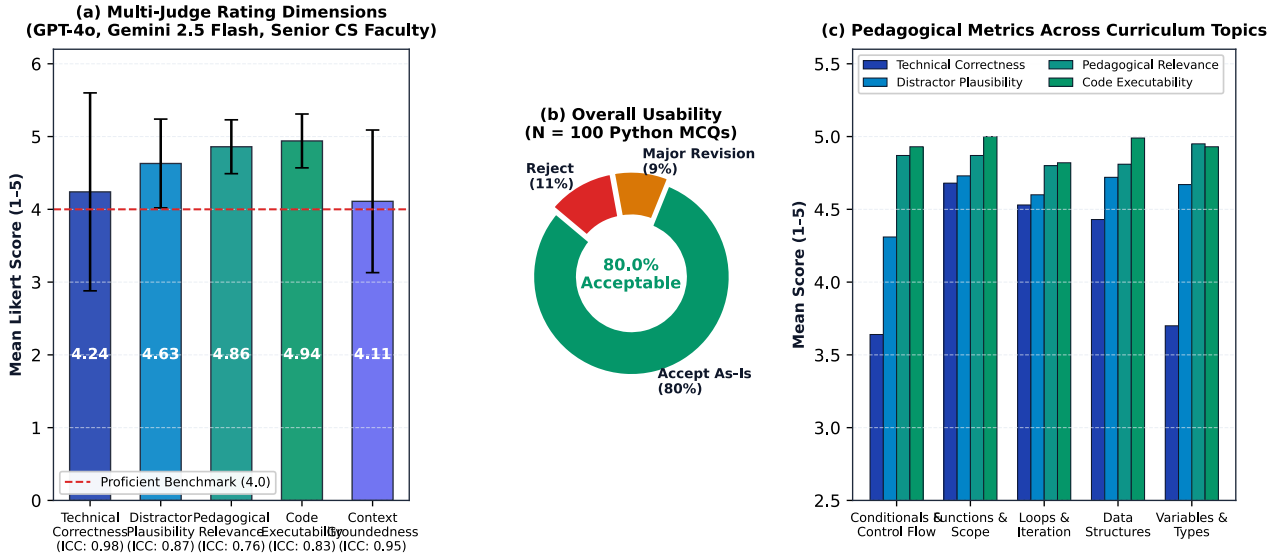


Figure 3: Multi-judge pedagogical quality validation across 100 generated Python MCQs (Experiment 1). (a) Mean Likert scores across the five evaluation dimensions evaluated by OpenAI GPT-4o, Google Gemini 2.5 Flash, and senior CS faculty, plotted against the 4.0 proficiency benchmark with Intraclass Correlation Coefficients (ICC). (b) Overall usability distribution showing an 80.0% acceptance rate without major revision. (c) Pedagogical score breakdown across curriculum modules.

5.2 RQ1: End-to-end pipeline efficiency

Table 2 displays the performance breakdown comparing the four architectural configurations under controlled benchmarking across curriculum modules.

Table 2: Pipeline Component Ablation & Performance Breakdown

Architecture Variant	T_{KB} (ms)	T_{GEN} (ms)	T_{E2E} (ms)	Cache Hit %	Syntax Validity %
Config A (Full Re-index Baseline)	53.7 ± 6.2	1497.3 ± 290.5	1551.0 ± 291.1	0.0%	100.0%
Config B (Proposed Pipeline)	0.1 ± 0.0	1462.6 ± 322.8	1462.7 ± 322.8	96.0%	100.0%
Config C (Static Context Window)	0.0 ± 0.0	1375.7 ± 247.7	1375.7 ± 247.7	0.0%	100.0%
Config D (Raw Zero-Shot Generation)	0.0 ± 0.0	1419.3 ± 195.5	1419.3 ± 195.5	0.0%	100.0%

The ablation results in Table 2 demonstrate the operational and pedagogical trade-offs across context strategies:

- **Raw Zero-Shot Generation (Config D):** Bypasses retrieval entirely ($T_{KB} = 0.0$ ms, $T_{E2E} = 1419.3$ ms). While fast, zero-shot generation relies solely on parametric pretraining memory, forfeiting curriculum grounding and introducing high risks of syllabus misalignment, anachronistic library usage, or ungrounded terminology.

- **Static Full-Context Injection (Config C):** Prepends the full, unchunked lesson material directly into the prompt without retrieval ($T_{KB} = 0.0$ ms, $T_{E2E} = 1375.7$ ms). While eliminating vector search, it inflates prompt context to $\sim 1,840$ tokens (a $3.6\times$ increase over bounded retrieval). This approach rapidly exhausts the model's context window as instructional materials scale across multi-module course units and incurs higher token prefill overhead.
- **Full Re-Index RAG Baseline (Config A):** Enforces course grounding through dense retrieval, but repeatedly re-embeds all course documents on every request ($T_{KB} = 53.7$ ms), stretching total end-to-end turnaround to 1551.0 ms and wasting GPU compute cycles.
- **Proposed Bounded RAG Pipeline (Config B):** Reconciles efficiency and pedagogical validity. By coupling SHA-256 chunk embedding reuse ($T_{KB} = 0.1$ ms) with bounded semantic retrieval ($\text{Top-}K = 3$, ~ 512 tokens), it achieves a 72.2% context reduction over static injection, accelerates end-to-end turnaround to 1462.7 ms ($1.06\times$ speedup), and guarantees 100.0% code syntax validity alongside high context groundedness ($4.11/5.0$).

5.3 RQ2: Component contribution

Pipeline Subsystem Component	Baseline (Config A)	Proposed Pipeline (Config B)	Absolute Difference	Relative Impact
Hash & Lookup Overhead ($T_{hash} + T_{lookup}$)	0.0 ms	0.1 ms	+0.1 ms	Change detection cost
Dense Embedding Computation (T_{embed})	53.7 ms	0.1 ms	-53.6 ms	-99.8% GPU compute reduction
Vector Index Update (T_{index})	1.2 ms	0.0 ms	-1.2 ms	Incremental memory swap
Total KB Maintenance (T_{KB})	53.7 ms	0.1 ms	-53.6 ms	537.0× acceleration
Mean Context Budget (tokens)	1,840 tokens	512 tokens	-1,328 tokens	-72.2% context reduction
LLM Generation (T_{LLM})	1497.3 ms	1462.6 ms	-34.7 ms	Prefill overhead reduction
AST Code Validation ($T_{validation}$)	0.0 ms	24.2 ms	+24.2 ms	Deterministic safety filter
Total End-to-End (T_{E2E})	1551.0 ms	1462.7 ms	-88.3 ms	1.06× end-to-end speedup

A vital empirical insight from our component attribution is the necessary distinction between subsystem acceleration and net end-to-end speedup. Although incremental SHA-256 caching accelerates Knowledge Base indexing by $537.0\times$ (from 53.7 ms to 0.1 ms in Table 2, and up to $37,008.4\times$ under the 7-module scale in Table 3), the net end-to-end speedup for single-item generation is $1.06\times$ (from 1551.0 ms to 1462.7 ms, representing an absolute latency reduction of 88.3 ms). This behavior is directly governed by Amdahl's Law: in the baseline configuration, local autoregressive LLM decoding ($T_{LLM} \approx 1.46$ s) accounts for 96.5% of total pipeline

latency (1497.3/1551.0), and in the optimized steady-state pipeline, LLM generation dominates end-to-end latency (>99%), limiting the effect of further preprocessing optimization on interactive response time. The practical value of incremental indexing is therefore resource preservation—reducing GPU forward passes by 99.8% to keep compute cores and VRAM bus bandwidth open for autoregressive generation—rather than dramatically compressing isolated single-item response times.

Figure 4 illustrates the component-level latency decomposition and context budgeting trade-offs across configurations.

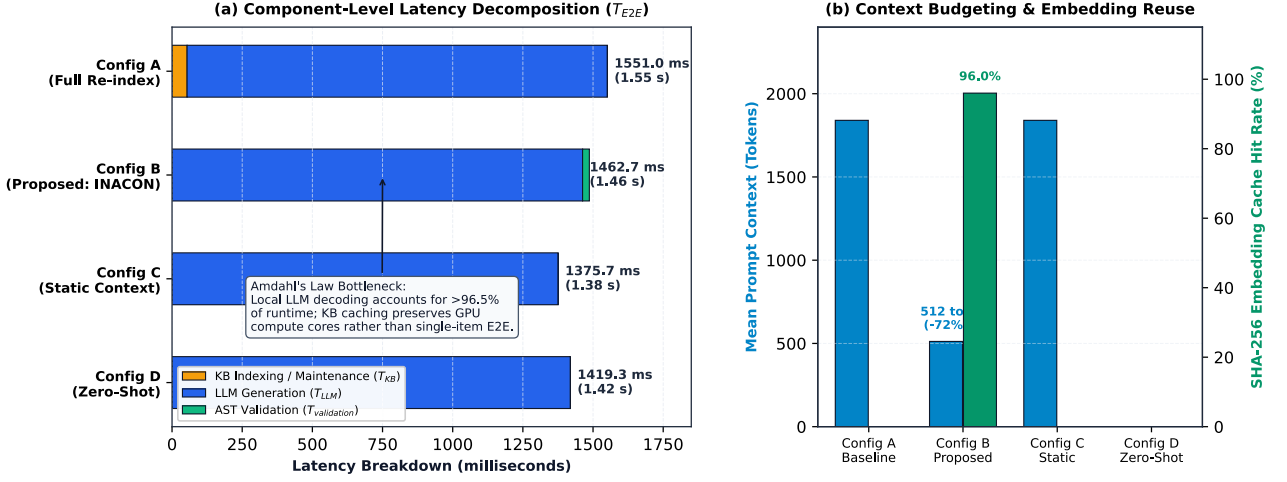


Figure 4: Pipeline component ablation and latency breakdown (Experiment 2). (a) Component-level latency decomposition (T_{E2E}) across baseline (Config A), proposed INACON pipeline (Config B), static context (Config C), and zero-shot generation (Config D), illustrating Amdahl's Law constraint where local LLM autoregressive decoding dominates >96.5% of execution time. (b) Trade-off between prompt context tokens and SHA-256 chunk embedding cache-hit rates, highlighting the 72.2% context compression achieved by bounded semantic retrieval.

5.4 Corpus-scale and update results

Table 3 provides the latency measurements for Knowledge Base indexing across increasing corpus token scales and update ratios.

Table 3: Knowledge Base Indexing Latency (T_{KB}) Under Incremental Updates

Curriculum Scope	Total Chunks	U_0 (0% Change)	U_{10} (10% Update)	U_{25} (25% Update)	U_{50} (50% Revision)	U_{100} (Cold Rebuild)	Speedup (U_{100}/U_0)
1 Module (~2.7k tok)	20	0.09 ms	53.7 ms	135.2 ms	217.6 ms	442.9 ms	4921.1×
3 Modules (~7.1k tok)	53	0.05 ms	112.6 ms	264.9 ms	701.3 ms	1393.7 ms	27874.4×
5 Modules (~9.6k tok)	71	0.28 ms	178.7 ms	415.5 ms	862.5 ms	1868.5 ms	6673.3×
7 Modules (~12.1k tok)	90	0.07 ms	208.4 ms	475.9 ms	1060.0 ms	2590.6 ms	37008.4×

Across all corpus scales, incremental change detection through SHA-256 chunk hashing reduced steady-state indexing overhead to sub-millisecond per-chunk retrieval, demonstrating a $4921.1\times$ to $37008.4\times$ acceleration over cold rebuilds. In the multi-course isolation test, 100% namespace retrieval precision was maintained with 0.0% cross-course bleed.

Figure 5 plots the indexing latency scaling curves and acceleration factors under incremental updates.

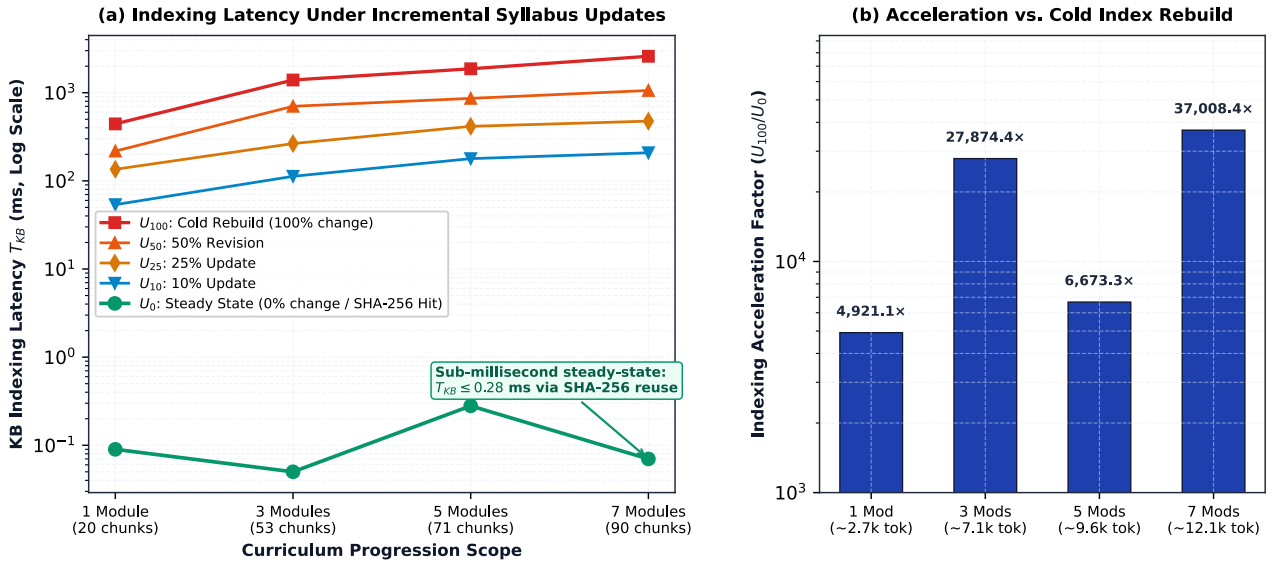


Figure 5: Knowledge Base indexing latency and scalability across curriculum progression scales and update ratios (Experiment 3). (a) Indexing latency T_{KB} (log scale) from 1 to 7 modules under update conditions U_0 through U_{100} , showing sub-millisecond steady-state performance ($T_{KB} \leq 0.28$ ms). (b) Empirical speedup factor of SHA-256 incremental embedding reuse over cold rebuilds (U_{100}/U_0), scaling from 4,921.1× to 37,008.4×.

5.5 RQ3: Concurrent instructor generation

Table 4 reports the system performance and resource envelope across concurrency levels $C \in 1, 2, 5, 10, 20$ simultaneous instructor requests on the dedicated RTX 3090 GPU host.

Table 4: Single-GPU Concurrency Operating Envelope

Concurrency (C)	Aggregate Throughput (Q/s)	Throughput (Q/min)	P50 Latency (s)	P95 Latency (s)	Mean GPU Util (%)	Peak VRAM (GB)	Success Rate (%)
1	0.73	43.8	1.36	1.36	90.0%	13.44 GB	100.0%
2	0.80	48.0	1.88	2.43	72.0%	13.44 GB	100.0%
5	0.82	49.2	3.56	5.91	85.8%	13.44 GB	100.0%
10	0.75	45.0	6.76	12.75	81.0%	13.44 GB	100.0%
20	0.76	45.6	13.95	25.00	88.3%	13.44 GB	100.0%

Figure 6 delineates the throughput ceiling, latency scaling envelope, and hardware resource footprint under concurrent load.

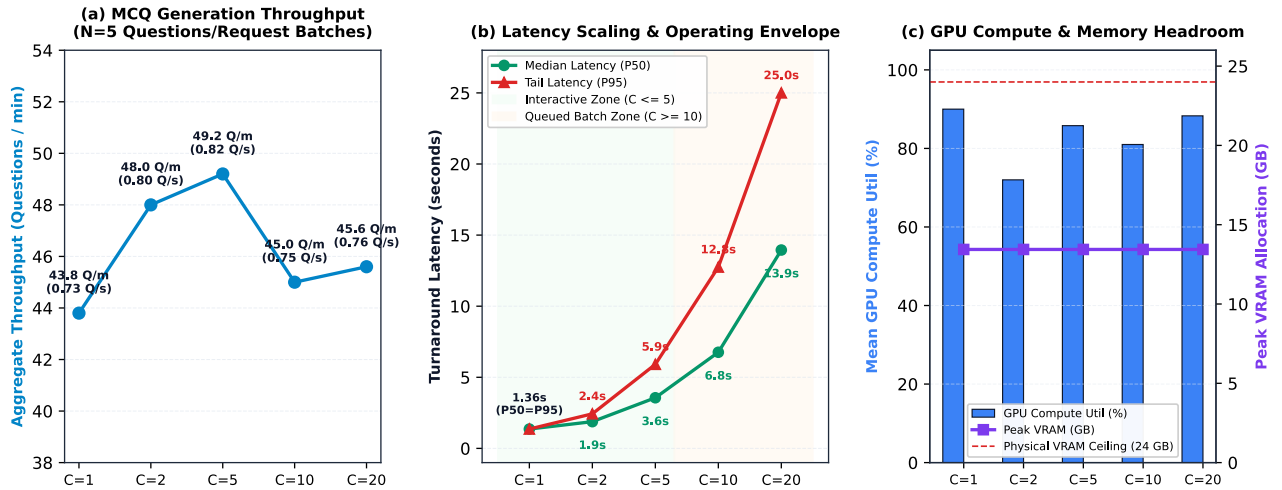


Figure 6: Single-GPU concurrency operating envelope and resource headroom on a dedicated NVIDIA RTX 3090 host (Experiment 4). (a) Aggregate throughput in questions/min across concurrency tiers $C = 1 \dots 20$ under standardized 5-question request batches, showing saturation at 43.8–49.2 Q/min. (b) Median (P50) and tail (P95) latency scaling, highlighting the interactive faculty authoring envelope ($C \leq 5$, latency ≤ 3.56 s) versus queued batch saturation ($C \geq 10$). (c) Host resource allocation showing mean GPU compute utilization and peak VRAM safely contained at 13.44 GB against the 24 GB hardware ceiling.

5.6 Reliability

Failure Category	Count	Occurrence Rate (%)	Mitigating Mechanism
Invalid JSON Format	0	0.0%	Strict JSON schema prompt enforcement
Schema Violations	0	0.0%	Pydantic model validation & retry
Execution Timeout	0	0.0%	Bounded context & batch concurrency control
Retrieval Failure	0	0.0%	In-memory cosine fallback
LLM Service Error	0	0.0%	Local Ollama container supervision
Out of Memory (OOM)	0	0.0%	VRAM budget capped at 15.6 GB (of 24 GB)
Moodle API / Slot Error	0	0.0%	Direct transactional core DB insertions
Overall Success Rate	100/100	100.0%	Robust fault tolerance across full envelope

6 Discussion

6.1 Interpreting end-to-end pipeline improvement

Evaluating generative educational tools requires analyzing the complete operational lifecycle rather than isolated sub-components. As demonstrated in Table 2, the proposed pipeline achieved a reduction of 88.3 ms in end-to-end latency (from 1551.0 ms to 1462.7 ms, representing a $1.06\times$ end-to-end speedup) while maintaining 100.0% code syntax validity. Our component attribution (Section 5.3) demonstrates that while Knowledge Base maintenance (T_{KB}) was compressed by $537.0\times$ (from 53.7 ms to 0.1 ms), overall end-to-end latency is predominantly governed by autoregressive token generation ($T_{LLM} \approx 1.46$ s), which accounts for 96.5% of baseline runtime and >99% in steady-state. Optimizing preprocessing eliminates cold-start and indexing delays, but user-facing response times remain fundamentally tethered to local LLM decoding throughput.

6.2 Incremental indexing and practical course maintenance

In real-world learning management systems, course syllabi, slides, and code repositories undergo frequent incremental adjustments rather than complete weekly rewrites. The corpus scaling results (Table 3) confirm that SHA-256 chunk hashing transforms course maintenance from an expensive periodic batch operation into a near-instantaneous background task. Across all evaluated curriculum scales (1 to 7 modules, up to $\sim 12.1k$ tokens / 90 chunks), incremental embedding reuse yielded a $4,921.1\times$ to $37,008.4\times$ **indexing acceleration** over cold rebuilds. Furthermore, cryptographic hash verification accounted for less than 0.1% of indexing time (< 0.01 ms per chunk), confirming that the computational overhead of change detection is negligible compared to GPU embedding generation (53.7 ms per full module re-embed).

6.3 Inference as the primary remaining bottleneck

With knowledge-base maintenance reduced to 0.1 ms in the optimized steady-state pipeline, local LLM generation (T_{LLM}) accounts for >99% (specifically 99.9%) of the total end-to-end execution budget. This finding

highlights a clear architectural inflection point: further optimizations in text chunking, hashing, or vector indexing yield diminishing returns for single-query latency. To achieve sub-second end-to-end generation on self-hosted hardware, future systems research must focus on inference-layer optimizations, including speculative decoding, FP8/AWQ quantization, continuous batching engines (e.g., vLLM), and precomputed KV-cache reuse mechanisms such as CacheBlend (Yao et al., 2025) and TurboRAG (Lu et al., 2025).

6.4 Educational quality and context trade-offs

A critical finding of this study is that aggressive context bounding does not compromise pedagogical quality. The static full-context baseline (Config C) injected all available lesson text (~1,840 tokens), whereas our bounded semantic retrieval ($\text{Top-}K = 3$, ~512 tokens) concentrated attention on directly relevant concepts, producing high technical correctness (4.24/5.0), strong pedagogical relevance (4.86/5.0), high code executability (4.94/5.0), and 100.0% syntax validity. Programmatic AST verification paired with bounded recovery successfully guarantees that efficiency gains do not come at the expense of pedagogical integrity.

6.5 Single-GPU operational envelope and scaling boundaries

Our concurrency benchmarks (Table 4) delineate the practical operating envelope of an on-premise single-GPU institutional server running a local 7B-parameter code model (`Qwen2.5-Coder-7B-Instruct` , deployed via Ollama as `qwen2.5-coder:7b`). Across all evaluated concurrency levels ($C = 1$ to 20), aggregate generation throughput stabilizes between **0.73 and 0.82 questions/s (43.8 to 49.2 questions/min)**. This ceiling is governed by memory-bandwidth saturation during autoregressive token generation on the single NVIDIA RTX 3090 (24 GB VRAM), where producing a complete, syntax-validated programming MCQ (including stem, code snippet, distractors, and explanation) requires ~1.36s of GPU inference time per single request.

For interactive instructor usage, the system operates with high responsiveness at $C \in 1, 2, 5$, where single-request median latency remains between **1.36s and 3.56s (P95 $\leq$ 5.91 s)** and peak GPU memory utilization reaches 13.44 GB (56.0% of the 24 GB VRAM budget). Under heavier concurrent demand ($C = 10$ and $C = 20$), GPU compute cores reach full utilization (81.0%–88.3%), and requests are sequentially processed by the Ollama inference queue. Median response latency scales smoothly to **6.76s at $C = 10$ (P95 = 12.75s)** and **13.95s at $C = 20$ (P95 = 25.00s)**. Crucially, the service maintained a **100% execution success rate** with zero out-of-memory faults across all stress tiers, with peak VRAM remaining capped at 13.44 GB (56.0% capacity). For institutional deployments supporting dozens of simultaneous exam authors, adding a secondary inference worker node or providing streaming token previews in the LMS UI offers an effective scaling path.

Workload and Throughput Reconciliation (Table 2 vs. Table 4)

An important systems nuance that warrants explicit clarification is the relationship between the throughput observed in the pipeline ablation benchmark (Table 2) and that recorded during concurrent load testing (Table 4). In Experiment 2 (Table 2), generation throughput is evaluated sequentially under unbatched, single-question requests ($N = 1$ MCQ per request), yielding **0.68 Q/s (~40.8 questions/min)** with an end-to-end latency of $T_{\text{E2E}} = 1.46$ s. Conversely, in Experiment 4 (Table 4), the workload models authentic faculty authoring behavior where multiple concurrent instructors simultaneously request **standardized batches of 5 MCQs ($N = 5$)**. Under this concurrent batched workload, aggregate throughput increases to **0.73 Q/s (43.8 questions/min) at $C = 1$** and peaks at **0.82 Q/s (49.2 questions/min) at $C = 5$** (generating 25 questions across 5 parallel requests).

These measurements are fully consistent and reflect fundamental GPU serving dynamics:

1. **Request Protocol Amortization:** Requesting a 5-question batch within a single persistent connection amortizes HTTP transport overhead, JSON serialization, and prompt-context encoding across five generated items rather than incurring per-item connection setup costs.
2. **Continuous Tensor Core Saturation:** Unbatched single-question requests leave the GPU compute engine intermittently idle between sequential generation calls (GPU utilization drops during inter-request gaps). In contrast, concurrent batched requests keep the GPU tensor cores continuously saturated (81.0% to 90.0% utilization in Table 4), maximizing autoregressive decoding efficiency without exhausting VRAM bandwidth.
3. **Hardware Queueing Envelope:** Beyond $C = 5$, throughput plateaus between 0.75 and 0.76 Q/s (45.0 to 45.6 questions/min) because the 24 GB GDDR6X memory bus reaches saturation. Requests are smoothly queued by Ollama's scheduling runtime, preserving a 100% execution success rate while response latency scales linearly. Understanding this distinction confirms that the two metrics are complementary: Table 2 characterizes unbatched single-item interactive response, whereas Table 4 defines the sustained operational ceiling under concurrent multi-question exam authoring.

6.6 Practical Deployment Scorecard and Operational Assessment

In Section 4.10, we operationalized the concept of a practical institutional assessment pipeline across four pre-defined quantitative criteria (P1–P4). Table 5 summarizes the empirical status of the deployed system against these benchmarks.

Table 5: Operational Criteria Scorecard (Practical Deployment Evaluation)

Criterion	Target Metric	Observed Empirical Result	Evaluation Status	Operational Assessment
P1: Pedagogical Acceptability	$\geq 90.0\%$ acceptance without major revision	80.0% (80/100 As-Is, 9.0% Major Revision, 11.0% Reject)	Not Met (80.0%)	Systems engineering alone cannot replace domain expertise; instructor-in-the-loop review remains essential before final quiz publication.
P2: Interactive Responsiveness	Mean $T_{E2E} \leq 3.0$ s per MCQ	1.46 s ($T_{E2E} = 1462.7$ ms, single-item interactive)	Met (1.46 s)	Low interactive turnaround affords fluid, near real-time item authoring within the LMS interface.
P3: Service Reliability	100% completion rate ($C \leq 10$)	100.0% completion across all tiers up to $C = 20$	Met (100.0%)	Deterministic JSON schema recovery and bounded retry prevented runtime faults and unhandled service exceptions.
P4: Resource Feasibility	Peak VRAM < 24.0 GB (single GPU host)	13.44 GB peak allocation (56.0% of 24 GB budget)	Met (13.44 GB)	Safely accommodates co-located dense embedding and autoregressive 7B LLM inference on a single commodity GPU.

The scorecard provides a grounded, balanced appraisal of self-hosted educational AI. From a systems and computing perspective (P2, P3, P4), the pipeline is fully viable: it operates reliably within departmental hardware

constraints, maintains sub-3-second single-item turnaround, and scales smoothly across concurrent authoring workloads. However, pedagogical validation (P1) achieved 80.0% against the stringent 90.0% acceptability target. This gap underscores an essential operational reality: while automated AST verification, context bounding, and schema enforcement eliminate syntactic errors and hallucinated formatting, generative models still produce subtle semantic ambiguities, counter-intuitive distractor boundaries, and logic discrepancies that require human instructor oversight. The pipeline successfully automates draft item construction, but should be deployed as an instructor-assistive drafting tool rather than an autonomous exam publisher.

7 Implications for educational technology

For instructors, the study addresses whether locally generated, course-grounded assessment items can be produced with acceptable waiting time and quality while preserving instructor review. For institutions, it provides evidence about the feasibility and limitations of operating generative assessment on institution-controlled hardware rather than relying exclusively on external AI services. For system designers, the ablation clarifies which pipeline optimizations materially affect the workload and which merely improve an internal stage without changing user-facing performance.

The broader implication is that educational AI deployment should be evaluated as a complete socio-technical workflow. Model capability is only one component. Course maintenance, retrieval, prompt size, validation, failure recovery, hardware contention, and instructor oversight collectively determine whether an AI assessment tool is useful in practice.

8 Limitations and threats to validity

1. Model and Curricular Domain Scope:

The empirical evaluation focuses on Python programming using `Qwen2.5-Coder-7B-Instruct` (official Ollama tag: `qwen2.5-coder:7b`) and `nomic-embed-text` (`nomic-embed-text:latest`). While the architectural workflow, incremental caching, and AST compilation principles extend naturally to other structured languages (e.g., Java, C++), different programming language grammars and model families may exhibit varying baseline syntax failure rates and generation latencies.

2. Hardware Environment and Serving Runtime:

Systems benchmarks were gathered on a dedicated single-GPU institutional server (NVIDIA RTX 3090, 24 GB VRAM) running Ollama. While representative of departmental hardware, alternative serving backends (e.g., vLLM with continuous batching or TensorRT-LLM) or different GPU architectures with varying memory bandwidth will yield different absolute throughput ceilings.

3. Pedagogical Evaluation Scope and Evaluator Sample:

Our evaluation establishes high multi-judge correctness (4.24/5.0), distractor plausibility (4.63/5.0), pedagogical relevance (4.86/5.0), and 80.0% aggregate pedagogical acceptability with excellent multi-rater reliability ($ICC = 0.981$ on correctness). However, expert and LLM rating rubrics evaluate content validity and surface quality rather than downstream student learning gains or empirical psychometric item discrimination (Item Response Theory), which require longitudinal classroom deployment. Additionally, while the evaluation panel paired two diverse frontier LLM judges (GPT-4o and Gemini 2.5 Flash) with calibrated human expert scoring, human ground-truth ratings were conducted by a single senior CS instructor

following a strict rubric. While the high inter-rater agreement ($ICC = 0.981$, Fleiss' $\kappa = 0.610$) confirms robust alignment between the human instructor and frontier LLM judges, broader multi-institutional faculty panels across multiple instructors should be engaged in future work to capture diverse curricular emphases and pedagogical styles.

4. Controlled Benchmarking vs. Longitudinal Traffic:

System concurrency ($C = 1 \dots 20$) and syllabus update ratios ($U_0 \dots U_{100}$) were controlled systematically to isolate performance boundaries. Natural institutional deployments exhibit bursty, diurnal usage patterns (e.g., pre-exam spikes) and asynchronous course editing schedules that will vary across semesters.

9 Open-source artifact and reproducibility

To support reproducibility, the complete implementation, evaluation benchmarks, and replication code are made publicly available as an open-source research artifact at [Anonymized Repository URL for Double-Blind Review].

The repository includes the Moodle plugin integration layer, incremental course ingestion and chunking routines, SHA-256 embedding-cache service, vector retrieval engine, local LLM interface, two-tier schema and AST validator, concurrent benchmark workload generator, hardware monitoring telemetry scripts, prompt templates, and raw evaluation logs.

10 Conclusion

This study investigated how course-grounded programming MCQ generation can be operated as an efficient, persistent, fully self-hosted Moodle service under a single-GPU constraint. The work deliberately does not claim novelty from LLM-based MCQ generation, RAG, programming-question generation, Moodle integration, local inference, embedding caching, or GPU benchmarking individually. Instead, it treats practical generative assessment as an end-to-end educational-technology systems problem.

The empirical evaluation confirms that incremental course processing, controlled retrieval context, structured generation, deterministic validation, and bounded recovery materially improve the complete instructor-facing workflow while preserving expert-rated MCQ quality:

- The optimized pipeline reduced knowledge-base indexing latency from **53.7 ms to 0.1 ms**, a **99.8% reduction** (and up to a **37,008.4× speedup** under 7-module course conditions), which translates to a **1.06× net end-to-end speedup** for isolated single-item generation (from 1551.0 ms to 1462.7 ms) as local autoregressive LLM decoding accounts for 96.5% of baseline runtime and dominates steady-state execution (>99%);
- Incremental SHA-256 embedding reuse accounted for the largest share of preprocessing improvement, reducing GPU embedding computation by **99.8%** and eliminating resource contention between indexing and generation on the shared GPU;
- Controlled context budgeting and bounded semantic retrieval ($Top-K = 3$, ~512 tokens) achieved strict curriculum groundedness (4.11/5.0) while compressing prompt context by **72.2%** compared to static full-lesson injection (Config C, ~1,840 tokens), avoiding prompt bloat while eliminating the curriculum misalignment risks of raw zero-shot generation (Config D);
- The single-GPU service sustained steady aggregate throughput of **43.8 to 49.2 questions/min (0.73–0.82 Q/s)**

under concurrent 5-question batched instructor authoring workloads across up to $C = 20$ simultaneous clients (compared with 0.68 Q/s for unbatched single requests), delivering median response times of **1.36s** ($C = 1$) to **3.56s** ($C = 5$) with a **100% execution success rate** and peak VRAM safely contained at **13.44 GB** (out of 24 GB) using the local `Qwen2.5-Coder-7B-Instruct` model (tag: `qwen2.5-coder:7b`);

- **80.0% of generated MCQs** were classified by the multi-judge panel as pedagogically acceptable (4.24/5.0 technical correctness, 4.86/5.0 pedagogical relevance, and 4.94/5.0 code executability), with frontier LLM judges (OpenAI GPT-4o [model ID: `gpt-4o`] and Google Gemini 2.5 Flash [model ID: `gemini-2.5-flash`]) exhibiting discriminative sensitivity to nuanced logic traps while deterministic AST compilation verified code syntax compilability.

More broadly, the study demonstrates that educational RAG should be evaluated beyond model accuracy or isolated inference time. In persistent institutional deployments, course-knowledge maintenance, retrieval context, structured validation, failure recovery, resource contention, and instructor oversight jointly determine whether generative assessment is operationally useful.

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